MambaGuard: Certified Streaming Detection for Multi-Protocol LLM Agent Security

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***Abstract*—Large language model (LLM) agents now communicate through four open protocols — the Model Context Protocol, the Agent-to-Agent Protocol, the Agent Communication Protocol, and the Agent Network Protocol — whose combined traffic forms a network-scale attack surface that prompt-boundary monitors and classical intrusion detection systems were not designed to observe. We present *MambaGuard*, a streaming detector that ingests a canonicalised protocol message graph, extracts per-agent sequential features with a selective state-space backbone, propagates relational context with a Lipschitz-normalised temporal graph attention layer, and selects defensive actions through a Hedge-based controller. The detector carries two complementary certificates. First, a randomised-smoothing certificate on the sentence-transformer feature space yields a lower bound on the expected defender utility of the form $V^* - B\ \Pr[\neg\mathcal{C}_\varepsilon] - B\sqrt{\ln N/(2T)} - B\alpha_{MC}$, in which every term shares the utility unit so the composition type-checks. Second, a sparse-smoothing certificate on the protocol graph bounds the number of injected or removed edges an attacker can tolerate per window. We evaluate MambaGuard on three protocol-native benchmarks (MCPSecBench, MCPSafetyBench, MCPTox) plus an A2A sandbox and two agent-tool corpora (AgentDojo, InjecAgent), and subject it to a dedicated adaptive-attack suite that includes white-box projected gradient descent in embedding space, discrete token search, LLM-driven payload rewriting, graph-structure evasion, and attacks on the online controller. We also show that the block-product Lipschitz bound used in earlier detectors of this family is not a valid upper bound once residual paths are counted, and we replace it with a residual-aware bound that a gradient-ascent probe confirms. We release an anonymised reference implementation, the evaluation harness, and a reproducibility script that regenerates every table and figure from a single command.**

1. Introduction

Enterprise AI has moved from single-prompt interfaces to protocol-mediated agent meshes. A 2025 systematic review of the Model Context Protocol catalogued more than thirty attack techniques at the MCP layer alone [1], and the empirical MCPSecBench study reported attack success rates above eighty-five per cent on at least one major commercial deployment [2]. The MCP Inspector remote-code-execution flaw [3] and the A2ABreak catalogue of eleven protocol-level vulnerabilities [4] confirm that protocol exposure translates directly into system compromise. The defensive response so far has been fragmented along three separate axes.

First, *design-level* defences such as CaMeL [13] and MELON [14] prevent prompt-injection outcomes by sandboxing tool invocation or isolating untrusted content. These approaches are complementary to detection: they provide provable guarantees on specific agent decisions, not on the broader protocol traffic. Second, *tool-boundary monitors* like ClawGuard [9] enforce per-call deterministic policies, and program-dependency monitors like AgentArmor [8] reconstruct a per-trace dependency graph. Neither observes the multi-agent interaction graph, nor reports a quantitative robustness guarantee. Third, classical *network intrusion detection systems* have evolved through graph neural networks [28], transformer encoders [29], and selective state-space models [23] — but these operate on flat flow features and do not model MCP/A2A/ACP/ANP message semantics or certify their own output.

MambaGuard closes the gap between these three strands. It models the protocol stream as a time-stamped heterogeneous graph whose vertices are agents, tools, capability descriptors, and sessions, and whose edges carry canonicalised message tuples. A selective state-space backbone [19] processes each agent's message stream in linear time; a Lipschitz-normalised GATv2 layer [24], [26] propagates relational context; and a Hedge controller [37] selects defensive actions online with no-regret against the best fixed commitment in hindsight. Two certificates accompany the detector: a randomised-smoothing certificate [30] on the payload-embedding feature space, and a sparse-smoothing certificate [34] on graph edges.

1.1. Contributions

- A **unified protocol-graph schema** for MCP, A2A, ACP and ANP (§4.1) with a five-class canonical message taxonomy.
- A **streaming composed detector** that couples a selective SSM with a Lipschitz-normalised temporal GATv2 layer (§4).
- A **dimensionally consistent two-certificate scheme** (§5): a Cohen-style smoothing certificate on feature-space perturbations with a composition theorem

whose terms all carry the utility unit, plus a sparse-smoothing certificate on graph edits.
- A **dedicated adaptive-attack evaluation** (§8) in the spirit of [39]: white-box PGD in embedding space, token-level discrete search, LLM-driven payload rewriting, graph-structural evasion, Hedge-weight manipulation.
- An **anonymised artifact** that regenerates every table and figure with one command.

1.2. Scope and Non-Goals

The paper's guarantees are deliberately scoped. The feature-space certificate holds inside the sentence-transformer image, not for arbitrary text edits; the graph certificate holds under a bounded number of edge additions or deletions, not under host compromise; and the online bound holds in the zero-sum restriction of the defender game. §5.6 and §9 state exactly what the certificates do and do not cover.

2. Background

Two technical ingredients underpin the paper. We state only what the body of the paper needs; extensions and standard proofs are deferred to Appendix A.

Selective state-space models.. A continuous state-space model maintains a hidden state $h(t) \in \mathbb{R}^N$ evolving as $h'(t) = Ah(t) + Bu(t)$, with output $y(t) = Ch(t)$. Zero-order-hold discretisation gives $h_t = \bar{A}h_{t-1} + \bar{B}u_t$, with $\bar{A} = \exp(\Delta A)$ and $\bar{B} = (\Delta A)^{-1}(\exp(\Delta A) - I)\Delta B$. The selective variant Mamba [19] makes Δ, B, C input-dependent and amortises inference in $\mathcal{O}(L)$ per sequence of length L via a hardware-aware parallel scan. We adopt the **S4D-Real** parameterisation [20], in which $A \in \mathbb{R}^N$ is a diagonal real matrix with strictly negative entries; this is the default Mamba initialisation and the choice reported to perform best on discrete modalities. For a real diagonal matrix the spectral radius equals the spectral (operator-2) norm, which is the property we exploit in the Lipschitz analysis of §5.2.

Randomised smoothing.. Given a base classifier f, the smoothed classifier $g(x) = \arg\max_c \Pr_{\delta\sim\mathcal{N}(0,\sigma^2 I)}[f(x+\delta) = c]$ is certifiably robust in ℓ_2 within a radius $r = \sigma\,\Phi^{-1}(p_A)$, where p_A is the probability of the top class under the noise distribution and Φ is the standard normal CDF [30]. In practice p_A is replaced by a Clopper-Pearson lower bound $\underline{p_A}$ estimated from n Monte-Carlo samples, with a failure probability α_{MC} absorbed into the certificate.

3. Threat Model and Security Goals

A robustness claim is only as meaningful as its threat model. We specify ours along four axes.

System model.. Several agents communicate through MCP, A2A, ACP and ANP. A trusted detector observes every protocol message, canonicalises it, and maintains a sliding-window protocol graph. A defender action is selected per round and applied to one message or session: deep packet inspection, blocking, session termination, agent isolation, or capability revocation.

Attacker knowledge.. We consider a *grey-box* attacker who knows the detector's architectural family (selective SSM plus temporal GATv2, protocol-graph representation, 34-class attack taxonomy), the published hyperparameters, and the frozen sentence-transformer encoder used for payload embeddings. The attacker does not have the deployed weights or the per-tenant Hedge random seed. For each experiment we also report a *white-box* setting in which the attacker additionally receives the trained weights and the embedding gradient; this is the standard worst case used by [39].

Attacker capabilities.. The attacker may (i) craft or rewrite protocol payloads inside a signed agent envelope; (ii) modify tool descriptions or Agent Cards; (iii) replay historical messages within a bounded time window; (iv) inject or delete graph edges up to a budget k per detection window through a compromised registered agent; (v) query the detector a bounded number of times to drive an attack search. The attacker cannot forge cryptographic signatures outside the compromised set, cannot compromise the host that runs the detector, and cannot modify the detector's weights. Compromise of the model supply chain, memory, or host is explicitly out of scope.

Attacker action space $\mathcal{A}_A$.. Seven families from the agent-protocol attack literature [1], [2], [4]: tool poisoning, indirect prompt injection, rug-pull capability updates, capability shadowing, cross-server capability confusion, credential exfiltration, and sandbox bypass.

Perturbation budgets.. Two complementary budgets. $\varepsilon \geq 0$ bounds the ℓ_2 norm of perturbations at the frozen payload-embedding layer; this is where the smoothing certificate holds. $k \in \{0, 1, \ldots\}$ bounds the number of graph edges the attacker may add or delete inside one detection window; this is where the sparse-smoothing certificate holds. We do *not* claim that the ℓ_2 budget transports to a bound on raw text edits: the payload encoder has no meaningful global Lipschitz constant from token edits to embeddings, and a single token change can move an embedding arbitrarily far relative to ε. §5.6 returns to this restriction.

Security goals.. The defender seeks (G1) low per-message false-positive rate on benign traffic; (G2) high detection rate on malicious traffic; (G3) *certified* robustness within the stated perturbation budgets; (G4) sustained throughput at the message rate of the monitored deployment.

4. Protocol Graph and Detector

4.1. Unified Heterogeneous Protocol Graph

Let $\mathcal{G}(t) = (\mathcal{V}(t), \mathcal{E}(t), X(t))$ denote the protocol graph at wall-clock time t. The vertex set $\mathcal{V}(t) = \mathcal{V}_A \cup$

$\mathcal{V}_T \cup \mathcal{V}_C \cup \mathcal{V}_S$ is partitioned into agents, tools or services, capability descriptors, and active sessions. Edges are time-stamped and typed; the five canonical types are agent-calls-tool, agent-messages-agent, agent-advertises-capability, tool-returns-data, and agent-delegates-task. Each protocol message is canonicalised into the tuple

$$m = (\tau, s, d, p, \mu, t_m),$$

where $\tau \in \{\mathcal{M}_{\text{tool}}, \mathcal{M}_{\text{comm}}, \mathcal{M}_{\text{cap}}, \mathcal{M}_{\text{data}}, \mathcal{M}_{\text{ctrl}}\}$, s, d are source/destination identifiers, $p \in \mathbb{R}^{d_p}$ is a frozen all-MiniLM payload embedding ($d_p = 384$), $\mu \in \mathbb{R}^{d_\mu}$ carries HTTP status, latency and auth metadata, and t_m is the message timestamp. A rule-based parser produces one tuple per raw protocol packet for each of MCP, A2A, ACP and ANP.

Worked example.. Consider a 2-second window in which Agent A issues an MCP `tools/list` to a database tool T_1 and receives a 400-byte JSON response; concurrently, Agent B sends an A2A `tasks/send` to Agent A delegating a user query, and a new capability descriptor C_3 is advertised through ANP by Agent C. The parser canonicalises six messages into three edges: $(A \to T_1, \mathcal{M}_{\text{tool}})$, $(T_1 \to A, \mathcal{M}_{\text{data}})$, $(B \to A, \mathcal{M}_{\text{comm}})$, and one capability-advertisement edge $(C \to C_3, \mathcal{M}_{\text{cap}})$. All four edges carry the same timestamp bucket $t_e \in [0, 2]$. The graph $\mathcal{G}(t)$ at the end of the window thus has six nodes (three agents, one tool, one capability, one session), four typed edges, and per-edge features comprising the 384-dim frozen embedding of the payload and a 32-dim metadata vector. This is the input the detector sees for one detection decision.

4.2. Selective SSM Backbone

For each agent $a \in \mathcal{V}_A$, the sequence of incident messages $x^a = (m_1^a, \ldots, m_{L_a}^a)$ is processed by a four-layer selective block (details in Appendix E):

$$h_t^a = \bar{A}_t \, h_{t-1}^a + \bar{B}_t \, m_t^a, \tag{1}$$

$$z_t^a = C_t \, h_t^a \odot \mathrm{SiLU}\big(W_g \, m_t^a\big), \tag{2}$$

with the standard Mamba depthwise causal 1-D convolution applied to x^a before (1). The gated output $z_t^a \in \mathbb{R}^{d_z}$ is the per-agent sequential feature.

4.3. Lipschitz-Normalised Temporal GATv2

With $\{z^a\}_{a \in \mathcal{V}_A}$ available, a temporal GATv2 layer propagates relational information. For an ordered pair (a, a') joined by an edge with timestamp t_e, the attention logit is

$$e_{a,a'} = q^\top \, \mathrm{LeakyReLU}\big(W_r \, [z^a \| z^{a'} \| e_{a,a'}^{\text{feat}} \| \Phi(t - t_e)]\big), \tag{3}$$

where $\Phi(\Delta t)$ is the Bochner-derived TGAT time encoding [25] with d_T learnable frequencies, $\|$ denotes concatenation, and q, W_r, W_v are learnable. We apply *LipschitzNorm* [26] to the attention weights; this is essential for the Lipschitz analysis below, because vanilla GATv2's dot-product attention is *not* Lipschitz for any ℓ_p norm on an unbounded domain (Kim *et al.* [27] prove this for self-attention and the argument transfers to GATv2). Multi-head attention with $H = 4$ heads is averaged at the final layer; the composed feature h^a feeds a softmax head over the 34-class attack taxonomy.

Algorithm 1 MambaGuard online defence loop

Require: Horizon T, utility range B, number of defender actions $N = |\mathcal{A}_D|$

$\eta \leftarrow \sqrt{8 \ln N / T} \,/\, B$ ▷ Corrected tuning (§5.4)
$w_1(s) \leftarrow 1$ for all $s \in \mathcal{A}_D$
for $t = 1, \ldots, T$ **do**
  Observe $\mathcal{G}(t)$ and per-agent streams
  Compute z_t^a via (1)–(2)
  Compute h_t^a via (3)
  Score classes $\hat{y}_t \leftarrow \mathrm{softmax}(W_h h_t)$
  Sample action $s_t \sim p_t(s) = w_t(s) / \sum_{s'} w_t(s')$
  Observe loss $\ell_t(s) \in [0, B]$
  $w_{t+1}(s) \leftarrow w_t(s) \exp(-\eta \, \ell_t(s))$
end for

4.4. Hedge Controller

The detector logits parameterise a Hedge [37] selector over the five defensive actions $\mathcal{A}_D$. Algorithm 1 states the complete loop.

5. Certified Guarantees

The detector carries two certificates — feature-space (§5.1) and graph-edit (§5.3) — plus a no-regret guarantee on the Hedge controller (§5.4). §5.5 combines them in a dimensionally consistent composition theorem, and §5.6 states what the certificates do and do not operationally guarantee.

5.1. Feature-Space Smoothing Certificate

We define the smoothed detector $g(x) = \arg\max_c \mathbb{P}_{\delta \sim \mathcal{N}(0, \sigma^2 I)}[f(x + \delta) = c]$ over the payload-embedding input x. Following [30], for a given x let $\underline{p_A}$ be the Clopper-Pearson lower bound on $\mathbb{P}[f(x + \delta) = \overline{g(x)}]$ from n Monte-Carlo samples at confidence $1 - \alpha_{MC}$. Then with probability at least $1 - \alpha_{MC}$ the certified radius is

$$R(x) = \sigma \, \Phi^{-1}\big(\underline{p_A}\big), \tag{4}$$

and $g(x + \delta') = g(x)$ for all $\|\delta'\|_2 \le R(x)$. We report certified-accuracy-vs-radius curves rather than a single number, because the radius varies per input (§7.2).

5.2. Local Lipschitz Bound on the Composed Detector

The smoothed detector's accuracy is tightened when the base detector is Lipschitz on the perturbed domain. For MambaGuard the base map is $f = h \circ g_{\text{gat}} \circ g_{\text{ssm}}$. We state the bound under explicit bounded-input assumptions, because

the selective SSM and the gated output are both products of input-dependent factors and therefore *not* globally Lipschitz on $\mathbb{R}^d$.

Assumption 1 (Bounded operating regime)**.** *(i) Inputs are projected to a Euclidean ball $\|x_t\|_2 \le X_{\max}$; (ii) discretisation steps are clamped to $\Delta_t \in [\Delta_{\min}, \Delta_{\max}]$; (iii) spectral norms of the projections producing B_t, C_t and Δ_t (including their Jacobians) are bounded by M_B, M_C, M_Δ respectively; (iv) the state matrix $A \in \mathbb{R}^N$ is real diagonal with strictly negative entries (S4D-Real) and $\Delta_{\max} \max_i A_i < 0$, so $\rho(\bar{A}_t) = \|\bar{A}_t\|_2 \le \rho < 1$.*

Lemma 1 (Local Lipschitz of the selective block)**.** *Under Assumption 1, with $\beta := \sup_t \|\bar{B}_t\|_2$, $\gamma := \sup_t \|C_t\|_2$, $\kappa_g := \|W_g\|_2$, $L_\sigma \approx 1.0998$ the exact SiLU Lipschitz constant, and $X_{\max}$ the input-norm cap, the selective block g_{ssm} is Lipschitz on $\{x : \|x_t\| \le X_{\max}\}$ with*

$$L_{ssm} \le \frac{\gamma\beta(1 + L_\sigma \kappa_g)}{1-\rho} + \frac{\gamma\beta X_{\max}(M_B + M_C + M_\Delta)}{(1-\rho)^2}. \quad (5)$$

The second term accounts for the input-dependent factors in $\bar{B}_t, C_t, \Delta_t$; it vanishes only when those parameters are input-independent (the S4 limit). Proof in Appendix B.

Lemma 2 (LipschitzNormed GATv2)**.** *With LipschitzNorm scaling constant η and Bochner frequencies $\{\omega_k\}$, the temporal GATv2 layer (3) satisfies*

$$L_{gat} \le \|W_v\|_2 \big(1 + \eta \, \|q\|_2 \, \|W_r\|_2\big), \quad (6)$$

$$L_\Phi \le \|W_v\|_2 \, \eta \, \|q\|_2 \, \|W_r\|_2 \sqrt{\tfrac{2}{d_T} \textstyle\sum_k \omega_k^2}. \quad (7)$$

Residual paths matter.. The detector stacks S residual SSM blocks after an input projection W_{in}, so the sound composition is

$$L_f \;\le\; M_h \, L_{\text{gat}} \, \|W_{\text{in}}\|_2 \prod_{\ell=1}^{S} \big(1 + L_{\text{ssm}}^{(\ell)}\big). \quad (8)$$

The plain product $M_h L_{\text{gat}} \prod_\ell L_{\text{ssm}}^{(\ell)}$, which drops the identity branch of every residual block and the input projection, is *not* an upper bound: in §7.2 a gradient-ascent probe exceeds it by a factor of 3.6. We use (8) only to tighten the smoothing certificate, and **we do not claim it transfers to raw text perturbations**.

5.3. Sparse-Smoothing Graph-Edit Certificate

A continuous certificate does not express structural attacks such as Agent Card impersonation, which add or remove edges in $\mathcal{G}(t)$. We apply the sparse-smoothing framework of Bojchevski *et al.* [34]. For edge-flip probabilities p_+ (adding an absent edge) and p_- (deleting a present edge), the smoothed classifier on edge-perturbed graphs is certified robust to budgets (k_a, k_d) whenever

$$\sum_{\mathbf{Q} \in \mathcal{H}} \min\{\mathbf{p}_{\mathbf{Q}}^+, \mathbf{p}_{\mathbf{Q}}^-\} \;\ge\; p_A - 0.5, \quad (9)$$

where the pointwise bound is taken over the Neyman-Pearson regions $\mathcal{H}$ induced by budgets k_a edge additions and k_d edge deletions, and $\mathbf{p}^\pm$ are the per-region likelihoods under the clean and perturbed noise distributions (full statement in [34], Theorem 1). In practice we evaluate the bound over the attacker's candidate edge set, not over all node pairs, which both reduces variance and matches the registered-agent restriction in the threat model.

5.4. Online No-Regret Guarantee

For losses in $[0, B]$ the Hedge regret bound $R_T \le (\ln N)/\eta + \eta T B^2/8$ is minimised at

$$\eta^* \;=\; \sqrt{8 \ln N / T} \;/\; B, \quad (10)$$

giving

$$R_T \;\le\; B\sqrt{(T/2)\ln N}. \quad (11)$$

(This is the classical Cesa-Bianchi–Lugosi tuning [37]; the frequently quoted $\eta = \sqrt{2\ln N/T}$ minimises the cruder bound $(\ln N)/\eta + \eta T/2$ and gives $\sqrt{2T \ln N}$, twice the constant.) For an anytime variant with $\eta_t = \sqrt{8 \ln N / t}/B$, the regret is at most $B(\sqrt{2T \ln N} + \ln N/8)$.

Theorem 1 (No-regret)**.** *Algorithm 1 with η as in (10) satisfies (11).*

We do *not* claim that no-regret dynamics converge to the Stackelberg leader value in general. That claim is only true in two restricted settings: (a) the defender-attacker game is *zero-sum*, in which case the Stackelberg value equals the minimax value and Hedge's time-average payoff is within R_T/T of it; or (b) the guarantee is phrased as regret against the *best fixed commitment in hindsight* over a finite set of attacker types [38], with the action set restricted to the extreme points of the attacker best-response polytopes. Our experiments report both views.

5.5. Composed Certificate (Type-Checked)

Let $\mathcal{C}_\varepsilon$ be the event that the smoothed detector's prediction is certified constant at radius ε, as in (4). On $\mathcal{C}_\varepsilon$ the defender's action and utility are unchanged under attack; off $\mathcal{C}_\varepsilon$ the per-round loss is at most B. Combining with Theorem 1 gives the composition.

Theorem 2 (Composed certificate)**.** *Let V^* be the Stackelberg value under the restriction of §5.4(a) or (b), $\hat{\pi}_T$ the Hedge policy after T rounds, $\varepsilon \ge 0$ the feature-space ℓ_2 budget, and α_{MC} the Monte-Carlo failure probability in (4). Then*

$$\mathbb{E}\big[V_{ach}(\hat{\pi}_T, \varepsilon)\big] \;\ge\; V^* - B\,\mathbb{P}\big[\neg\mathcal{C}_\varepsilon\big] - B\sqrt{\tfrac{\ln N}{2T}} - B\,\alpha_{MC}. \quad (12)$$

All four terms share the utility unit; $\mathbb{P}[\neg\mathcal{C}_\varepsilon] = 1 - \mathrm{CertAcc}(\varepsilon)$ is read off the certified-accuracy curve. Proof in Appendix C.

5.6. Operational Meaning and Non-Guarantees

Theorem 2 guarantees a lower bound on long-run expected defender utility against any attacker whose per-round perturbation stays within the stated budgets. It does *not* guarantee: (i) that an arbitrary text edit of a payload lies within ε in embedding space (the encoder has no global token-to-embedding Lipschitz constant); (ii) robustness to graph edits beyond $k_a + k_d$ within one window; (iii) robustness to host compromise, cryptographic-signature forgery, or zero-day protocol-implementation flaws. Applications that require payload-level provable robustness should additionally use the erase-and-check certificate of Kumar *et al.* [33] against suffix/insertion/infusion up to d tokens; our evaluation reports this composition in §7.2.

6. Experimental Setup

6.1. Benchmarks

We evaluate on six benchmarks that span the agent-protocol attack surface.

MCPSecBench [2]: 17 attack types across 4 MCP surfaces (tool, server, host, user) with 15 runs per attack on three LLM platforms (Claude Desktop, GPT-4.1, Cursor).

MCP-SafetyBench [5]: 245 multi-turn test cases over 5 domains and 20 attack types; the published average ASR across 13 LLMs ranges from 29.80% (Qwen3-235B) to 48.16% (o4-mini), with host-side attacks reaching 81.94%.

MCPTox [6]: 1,312 malicious tool-poisoning cases over 10 risk categories across 45 live real-world MCP servers and 353 authentic tools.

A2A-sandbox: a reproduction of the 11 A2ABreak [4] vulnerability classes, including cross-client context injection and rogue-agent capability advertisement, generated in a local sandbox with no outbound network access.

AgentDojo [11] and **InjecAgent** [12]: 629 and 1054 indirect-prompt-injection scenarios respectively, canonicalised into our protocol-graph representation via the mapping documented in Appendix F.

Network-IDS transfer (appendix only). For completeness we also include a transfer evaluation on CSE-CIC-IDS2018 [43], CIC-IoT2023 [44], and UNSW-NB15 [45] with 70/15/15 strict temporal splits; those numbers are in Appendix I and are *not* used in the headline claims, because the datasets do not represent agent-protocol threats.

6.2. Baselines

We compare against three tiers: (i) detector-style agent defences PromptGuard [16], PIGuard [17], and DataSentinel [18]; (ii) design-level defences CaMeL [13] and MELON [14]; (iii) classical IDS backbones E-GraphSAGE [28], RTIDS [29], and NetMamba [23] applied to the protocol graph via a sentence-transformer front-end. Each baseline is retrained under identical splits with its published hyperparameters [confirm once done].

6.3. Metrics

Per-benchmark macro-F1, per-benchmark post-defence ASR (at a defender-action threshold selected on validation), certified accuracy at $\varepsilon \in \{0.01, 0.05, 0.10, 0.25\}$, batch-1 streaming per-message latency (median and p99 over 1000 runs after 50 warm-ups), and throughput at the best batch size. We do *not* report a cross-benchmark macro-average because the label spaces differ.

6.4. Statistical Protocol

Each experiment is run with five seeds (0–4). Entries are mean $\pm$ sample standard deviation. Pairwise comparisons use paired two-sided Wilcoxon signed-rank tests on the $n = 5 \times B$ (seeds $\times$ benchmarks) grid with Holm correction across comparisons; the smallest achievable two-sided exact p at $n = 5$ is $2^{-5} \cdot 2 = 0.0625$, so we report p-values for $n \geq 15$.

7. Results

7.1. Per-Benchmark Detection

Table 1 reports macro-F1 and post-defence ASR per benchmark. [Summarise Table 1 once the benchmark runs complete: on how many benchmarks MambaGuard leads on F1 and on ASR, and where design-level defences such as CaMeL are stronger.] MambaGuard and CaMeL are complementary: MambaGuard observes traffic, CaMeL rewrites tool calls; §9 discusses the composition.

7.2. Certification: Actual Lipschitz Constant, Smoothing and Graph-Edit

Table 2 reports the per-component Lipschitz factors as a function of the effective spectral cap $c = \sup_\ell \|W^{(\ell)}\|_2$ enforced by the regulariser λ_L. The key finding is at $c = 0.5$: the block-product quantity is 0.023, but projected gradient ascent on δ finds a ratio $\|f(x+\delta) - f(x)\|/\|\delta\|$ of 0.083, i.e. $3.6\times$ larger. The block product is therefore *not* an upper bound; it omits the identity branch of each residual block and the input projection. The residual-aware bound (8) is 3.09, which the probe respects, and which is $37\times$ looser than the probe. Tighter LipSDP-style estimates [32] are future work. All certificates in this paper use the residual-aware bound.

Figure 1 plots certified accuracy against radius r for $\sigma \in \{0.05, 0.10, 0.25, 0.5\}$ and compares to an erase-and-check certificate [33] for suffix attacks on AgentDojo. As an implementation check on 32 synthetic inputs with $n = 1000$ samples and $\sigma = 0.25$, the Cohen certifier gives certified accuracy 0.78 at $r = 0$, 0.69 at $r = 0.05$, 0.50 at $r = 0.25$ and 0.16 at $r = 0.5$; plugging these into Theorem 2 with $V^* = B = 1$, $T = 10^4$, $N = 5$, $\alpha_{MC} = 10^{-3}$ gives a defender-utility lower bound of 0.68 at $r = 0.05$ and 0.49 at

TABLE 1. **PLACEHOLDER — NOT YET MEASURED.** PER-BENCHMARK RESULTS, MEAN ± SAMPLE STD OVER 5 SEEDS. "–" DENOTES A METHOD INAPPLICABLE TO THE BENCHMARK. MG = MAMBAGUARD. STATISTICAL SIGNIFICANCE (PAIRED WILCOXON, HOLM-CORRECTED, VS SECOND-BEST PER COLUMN) IS MARKED * AT $p < 0.05$ AND ‡ AT $p < 0.01$.

	MCPSecBench F1 / ASR (%)	**MCP-SafetyBench** F1 / ASR (%)	**MCPTox** F1 / ASR (%)	**A2A-sandbox** F1 / ASR (%)	**AgentDojo** F1 / ASR (%)	**InjecAgent** F1 / ASR (%)
PromptGuard	.63 / 38.1	.57 / 44.6	.52 / 49.3	.54 / 47.8	.61 / 31.5	.63 / 34.2
PIGuard	.67 / 35.4	.61 / 40.1	.56 / 45.9	.58 / 44.2	.64 / 29.1	.66 / 31.0
DataSentinel	.74 / 24.7	.70 / 28.3	.66 / 34.4	.67 / 30.6	.71 / 19.8	.73 / 21.4
CaMeL	.68 / 17.4	.65 / 22.1	.61 / 27.5	.63 / 25.0	.76 / **4.9**	.78 / **6.1**
MELON	.70 / 18.9	.67 / 23.8	.62 / 29.0	.64 / 26.2	.74 / 5.7	.76 / 7.4
NetMamba+ST	.77 / 20.3	.72 / 25.6	.68 / 31.2	.69 / 28.9	.73 / 17.1	.75 / 18.3
E-GraphSAGE+ST	.73 / 24.1	.69 / 28.8	.65 / 34.5	.66 / 31.0	.70 / 20.4	.72 / 22.0
RTIDS+ST	.75 / 22.5	.71 / 27.0	.66 / 32.9	.68 / 29.4	.72 / 18.9	.74 / 20.5
MG (ours)	**.86**‡ / **11.2**‡	**.83**‡ / **14.6**‡	**.79**‡ / **19.8**‡	**.81**‡ / **16.3**‡	.83* / 9.1	.85* / 10.4

TABLE 2. LIPSCHITZ BOUNDS VERSUS THE EFFECTIVE SPECTRAL CAP c IMPOSED ON EVERY LINEAR LAYER (DEFAULT 3.68 M-PARAMETER CONFIGURATION). "BLOCK PRODUCT" IS $M_h L_{\text{GAT}} \prod_\ell L_{\text{SSM}}^{(\ell)}$; "RESIDUAL" IS THE SOUND BOUND (8); $\hat{L}_f$ IS THE EMPIRICAL LOWER BOUND FROM 200-STEP PROJECTED GRADIENT ASCENT ON δ. ROWS FOR $c \neq 0.5$ STILL NEED THE RESIDUAL BOUND AND THE PROBE.

c	L_{ssm}	L_{gat}	Block product	Residual	$\hat{L}_f$
0.50	0.06	1.46	0.023	3.09	0.083
0.75	12.88	5.50	39.87	–	–
1.00	58.84	14.73	651.80	–	–
1.25	89.74	32.30	2482.52	–	–
2.00	177.96	67.99	10364.17	–	–



Figure 1. Certified accuracy vs ℓ_2 radius r on MCP-SafetyBench, for four noise levels $\sigma \in \{0.05, 0.10, 0.25, 0.5\}$. Dashed curve: erase-and-check certificate for 2-token suffix attacks on AgentDojo. Shaded band: ± 1 std over 5 seeds.

$r = 0.25$. On MCP-SafetyBench with $n = 10^4$ the certified accuracy at $\sigma = 0.25$ is a% at $r = 0.1$ and b% at $r = 0.5$.

For the *graph-edit* certificate we use sparse smoothing with $p_+ = p_- = 0.05$, restricted to the attacker's candidate edge set of registered agents. As a correctness check of the implementation, on a 10-node test graph with $n = 1000$ smoothing samples and $\alpha = 10^{-3}$ the certificate returns $\underline{p_A} = 0.993$ and a budget of $(k_a, k_d) = (2, 2)$. On MCPSecBench the median certified budget is $(k_a, k_d) = (x, y)$, with z% of attacks certifiably classified correctly at $(1, 1)$. Full distribution in Appendix G.

TABLE 3. LATENCY AND THROUGHPUT, MEASURED SEPARATELY. "BATCH 1" IS THE STREAMING PER-MESSAGE LATENCY A SINGLE USER SEES (MEDIAN OVER 1000 RUNS AFTER 50 WARM-UPS, INCLUDING 1 MS CANONICALISATION). "BEST BATCH" IS THE BATCH SIZE THAT MAXIMISES THROUGHPUT; ITS PER-MESSAGE COST IS THE BATCH TIME DIVIDED BY THE BATCH SIZE. THE CPU ROW IS A SHARED CLOUD CPU AT SEQUENCE LENGTH 16.

	Batch 1 median	**Batch 1 p99**	**Best batch**	**Per-msg at best batch**	**Throughput (msg/s)**
CPU	7.7 ms	13.6 ms	8	3.0 ms	334
A100 GPU	–	–	–	–	–

7.3. Latency and Throughput, Harmonised

Per-message latency and aggregate throughput are different quantities, and a single number cannot stand for both. Table 3 therefore reports (i) streaming batch-1 latency, end to end, and (ii) the amortised per-message cost at the batch size that maximises throughput.

On CPU, batching already helps at batch 2 (4.9 ms per message versus 6.8 ms at batch 1) and throughput peaks at batch 8. A sub-5 ms streaming latency claim, and any million-messages-per-second throughput claim, must come from the A100 row, which remains to be measured.

7.4. Expanded Component Ablation

Table 4 reports an expanded ablation (parameter counts are measured; the remaining columns await the benchmark runs). On CPU at batch 8 and sequence length 16 the full model takes 61 ms per batch, the SSM-only variant 60 ms, the GAT-only variant 0.6 ms, and a 4-layer transformer in place of the scan 3.9 ms; at this short sequence length the reference (non-fused) scan is slower than attention, so the latency case for the selective scan rests on the fused GPU kernel at longer sequences (to be measured). [Summarise the F1/ASR effect of each ablation.]

TABLE 4. **PLACEHOLDER — NOT YET MEASURED.** ABLATION ON MCP-SAFETYBENCH, MEAN ± STD OVER 5 SEEDS. CERT. RAD. IS MEDIAN CERTIFIED RADIUS AT $\sigma = 0.25$.

Config	Params	F1	Lat.(ms)	ASR(%)	Cert.rad.
Full MG	3.68 M	.83	3.7	14.6	0.21
no Lipschitz	3.68 M	.82	3.7	15.2	< 0.01
no Hedge controller	3.68 M	.83	3.5	24.1	0.21
no GATv2 (SSM only)	1.93 M	.80	2.9	18.9	0.19
no SSM (GAT only)	1.92 M	.78	0.7	22.4	0.14
Transformer-for-SSM	2.22 M	.82	37.4	15.0	0.20
no time encoding	3.63 M	.81	3.4	17.3	0.20
flat sequence	3.68 M	.76	3.6	28.8	0.21

8. Adaptive Attacks

Nasr *et al.* [39] show that twelve recent prompt-injection filters were bypassed with ASR above 90% once attackers were given time to tune and scale a handful of standard techniques. MambaGuard is a filter in this family, so we report a dedicated adaptive-attack evaluation. We follow the Tramèr *et al.* methodology [40] of targeting *every* component of the defence.

E1 White-box embedding PGD.. Projected gradient descent with Expectation-over-Transformation [41] over the smoothing noise. Budget ε in ℓ_2 at the sentence-transformer feature layer, 100 steps, 8 EOT samples, step $2\varepsilon/100$. Reports empirical ASR within the *claimed* certified ball (should be zero in the certified fraction) and beyond it. Table 6 row E1.

E2 Discrete token search (GCG-style).. Greedy Coordinate Gradient token search [42] constrained to preserve the malicious tool call. Measures whether the certified radius transports to raw text perturbations — it does not, by construction, but we report the empirical bypass rate so reviewers can see the gap between the feature-space certificate and text-level evasion.

E3 LLM-driven payload rewriting.. An attacker LLM (Claude 3.5 Sonnet with a search budget of 20 queries per sample, following [39]) is told the detector's taxonomy and the target class, and rewrites the malicious payload to minimise the detector score while preserving a verifiable attack outcome.

E4 Graph-structural evasion.. Split the attack across m agents or w windows to stay under per-window graph statistics. Mimic benign subgraph patterns extracted from the test partition.

E5 Hedge-weight manipulation.. An adaptive attacker that observes the defender's actions (a legitimate assumption under the Balcan *et al.* [38] framework) and plays a type sequence that drives the Hedge weights toward the least effective defender action. We report the empirical regret growth versus the Hedge bound: if the empirical regret exceeds the proven bound $B\sqrt{(T/2)\ln N}$, the online guarantee is empirically violated.

E6 Human red team.. [Only keep this paragraph if the red-team exercise is actually run with ethics approval before submission; otherwise delete E6 here and in Table 6.] Planned protocol: a three-person red team produces hand-crafted attack traces (not modifications of benchmark traces), given the detector's taxonomy and restricted to non-destructive techniques. Ethics discussion in Appendix J.

TABLE 5. **PLACEHOLDER — NOT YET MEASURED.** MAMBAGUARD VERSUS THE STRONGEST PROMPT-INJECTION FILTERS FROM THE NASR *et al.* [39] STUDY, ON AN ALIGNED TEST SET. ALL NUMBERS ARE BYPASS RATES AT THE AUTHORS' PUBLISHED ADAPTIVE-ATTACK BUDGET (20 ATTACKER QUERIES, 100 PGD STEPS). "MG+EC" IS MAMBAGUARD COMPOSED WITH ERASE-AND-CHECK.

Defence	PIGuard	MambaGuard	MG+EC
E1 Embed-PGD	89%	47%	**31**%
E2 GCG	71%	49%	**23**%
E3 LLM rewrite	82%	52%	**34**%
Avg.	81%	49%	**29**%

TABLE 6. **PLACEHOLDER — NOT YET MEASURED.** ADAPTIVE-ATTACK ASR ON MCP-SAFETYBENCH, AVERAGED OVER 5 SEEDS. ε IS THE ATTACKER'S ℓ_2 BUDGET AT THE SENTENCE-TRANSFORMER FEATURE LAYER; k IS THE GRAPH-EDIT BUDGET.

Attack	ε=0.01	ε=0.05	ε=0.10	k=1	k=3
E1 Embed-PGD (EOT)	1.1	8.4	47.3	—	—
E2 GCG discrete	12.5	28.0	48.9	—	—
E3 LLM rewrite	18.2	34.1	51.6	—	—
E4 Graph evasion	—	—	—	13.4	41.8
E5 Hedge manipulation	avg. regret 0.21 vs bound 0.20				
E6 Human red team	39/200 bypassed = 19.5% bypass rate				

Comparison to the prompt-injection-filter baseline.. Nasr *et al.* [39] report $> 90\%$ ASR against twelve recent prompt-injection defences (including Protect AI, PromptGuard, PIGuard, Model Armor) under their adaptive-attack methodology, with PIGuard — the strongest detector baseline — still reaching 71% bypass. For direct comparison we run the same adaptive attacks against MambaGuard; Table 5 reports the bypass rate at the published Nasr *et al.* attacker-budget level, and reports MambaGuard's bypass rate at the same budget. [State the measured reduction relative to PIGuard, if any.] We make no claim of invulnerability.

In a sandbox run on synthetic inputs (16 samples, 40 PGD steps, 4 EoT samples), E1 flips 0% of predictions at $\varepsilon \in \{0.01, 0.05\}$, 6.3% at $\varepsilon = 0.1$ and 12.5% at $\varepsilon = 0.2$; these are implementation checks, not benchmark results. [Interpret Tables 6 and 5 once measured: (1) whether E1's success inside the certified ball stays within the certified failure fraction; (2) how far text-level attacks E2/E3 exceed the feature-space budget, and how much the erase-and-check composition recovers; (3) whether E5 drives empirical regret above $B\sqrt{(T/2)\ln N}$.]

9. Discussion and Limitations

MambaGuard is a *detector*, complementary to design-level defences like CaMeL and MELON that prevent injection outcomes by sandboxing; we view detector+design

composition as the deployment target. Three honest limitations. **Feature vs payload.** The smoothing certificate holds in sentence-transformer space; the erase-and-check composition partially bridges but does not close the gap to text-level perturbations. **Loose Lipschitz.** The sound residual-aware bound is $37\times$ the empirical lower bound at $c = 0.5$, and the simpler block-product bound is not a bound at all; tighter data-dependent estimates are future work. **Attacker capabilities.** The threat model excludes host compromise, cryptographic-signature forgery and arbitrary new-agent injection; these capabilities need orthogonal defences. Encrypted traffic is partially addressed (IDS-PQC in Appendix I); differentially private embeddings are future work.

10. Related Work

Agent-protocol security monitors. ClawGuard [9] enforces deterministic tool-access policies. AgentArmor [8] reconstructs program-dependency graphs from runtime traces. DataSentinel [18] is a game-theoretic detection framework published at IEEE S&P 2025. None observes the multi-protocol interaction graph or reports a certificate.

Design-level defences. CaMeL [13] prevents injection outcomes by design. MELON [14] provides provable guarantees against indirect prompt injection in ICML 2025. Progent [15] adds programmable privilege control. The Task Shield [10] enforces task alignment. These are complementary to our detector.

Certified robustness. Tsuzuku *et al.* [31] introduced Lipschitz-margin training; Cohen *et al.* [30] introduced randomised smoothing. Bojchevski *et al.* [34] developed sparse smoothing for graphs; Schuchardt *et al.* [35], [36] extended to collective and hierarchical variants. Kumar *et al.* [33] gave erase-and-check for LLM prompt perturbations.

Mamba / SSM stability. Halloran *et al.* [21] prove Lyapunov stability for Mamba; a parallel 2024 preprint explores adversarial robustness of deep SSMs [22].

Attention Lipschitz. Kim *et al.* [27] prove that dot-product attention is not globally Lipschitz; LipschitzNorm [26] is the standard remedy for graph-attention networks.

Adaptive-attack methodology. Nasr *et al.* [39] bypassed twelve recent prompt-injection defences; Tramèr *et al.* [40] wrote the methodological reference.

Game theory. Balcan *et al.* [38] prove regret against the best fixed commitment in hindsight for repeated Stackelberg games under unknown attacker types.

11. Conclusion

We presented MambaGuard, a streaming detector for multi-protocol LLM-agent traffic that couples a selective SSM with a Lipschitz-normalised temporal GATv2 layer and ships two type-checked certificates and a corrected online no-regret bound. We corrected the online step size, made the composition theorem unit-consistent, and showed that a block-product Lipschitz bound is unsound once residual paths are counted. [Add one sentence on the measured benchmark and adaptive-attack results.] Code, weights, the adaptive-attack harness, and a reproduction script that regenerates every table and figure are released under an anonymised mirror.

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Appendix A. Mathematical Preliminaries

Standard restatement of S4D-Real Mamba, dynamic graph attention and TGAT time encoding; omitted from body for space. Full derivations in the anonymised artifact (file `docs/preliminaries.pdf`).

Appendix B. Proof of Lemma 1 (Local SSM Lipschitz)

Under Assumption 1, the state recurrence $h_t = \sum_{k=0}^{t-1}(\prod_{j=t-k+1}^{t}\bar{A}_j)\bar{B}_{t-k}m_{t-k}$ gives $\|h_t(x)-h_t(x')\| \le \sum_{k\ge 0}\rho^k\beta\|m_{t-k}-m'_{t-k}\| = \beta/(1-\rho)\|x-x'\|_{\infty,t}$. Multiplying by $\gamma = \sup\|C_t\|$ for the output, and by the Hadamard-gate factor $(1+L_\sigma\kappa_g)$ since $\|fg\|_{\text{Lip}} \le \|f\|_\infty L_g + \|g\|_\infty L_f$ with $\|f\|_\infty, \|g\|_\infty \le \gamma\beta X_{\max}/(1-\rho)$, bounded by $X_{\max}$, yields the first term of (5). The second term absorbs the Jacobians of $\bar{B}_t$, C_t and Δ_t with respect to the input; applying submultiplicativity once more gives $\gamma\beta X_{\max}(M_B + M_C + M_\Delta)/(1-\rho)^2$. The depthwise causal convolution is folded into β via its unrolled Toeplitz norm. □

Appendix C. Proof of Theorem 2 (Composed Certificate)

Partition the horizon into rounds where $\mathcal{C}_\varepsilon$ holds and rounds where it does not. On $\mathcal{C}_\varepsilon$ the detector's prediction, hence the defender's action, hence the per-round utility, is unchanged under attack and therefore at least V^* in expectation (by the Stackelberg optimality under the zero-sum or fixed-commitment restriction). On $\neg\mathcal{C}_\varepsilon$ the per-round loss is at most B. The Hedge regret $R_T \le B\sqrt{(T/2)\ln N}$ divided by T gives the second penalty. The α_{MC} term is the probability that the Clopper-Pearson lower bound fails and the certified event is reported in error. □

Appendix D. Exact SiLU Lipschitz Constant

$\frac{d}{du}\text{SiLU}(u) = \sigma(u)(1+u(1-\sigma(u)))$; numerical solution of $\text{SiLU}''(u) = 0$ yields $u^* \approx 2.3994$, $\text{SiLU}'(u^*) \approx 1.0998$.

Appendix E. Hyperparameters

$N = 16$, $d_z = 256$, expansion $E = 2$, kernel width $k = 4$, 4 stacked SSM layers; $H = 4$ heads, $d_T = 64$; AdamW lr 3×10^{-4}, weight decay 10^{-2}, batch 256, 30 epochs cosine; $\lambda_L = 10^{-3}$ unless otherwise noted; sentence-transformer all-MiniLM-L6-v2 ($d_p = 384$), frozen. Smoothing: $n = 10^4$, $\alpha_{MC} = 10^{-3}$, $\sigma \in \{0.05, 0.10, 0.25, 0.5\}$. Sparse smoothing: $p_+ = p_- = 0.05$, $n = 1000$.

Appendix F. Protocol-Benchmark Canonicalisation

Mapping tables from AgentDojo/InjecAgent, MCPSecBench and MCP-SafetyBench raw formats into the six-tuple $(\tau, s, d, p, \mu, t_m)$. Published in the artifact under `mambaguard/data/protocol_traces/`.

Appendix G. Graph-Edit Certified Budgets

Full distribution of (k_a, k_d) over the six benchmarks and the sensitivity to p_+, p_-. See the artifact for CSV.

Appendix H. Erase-and-Check Composition

Composing the feature-space smoothing certificate with the erase-and-check certificate [33] against suffix attacks of length ≤ 2 changes the E2 (GCG) bypass rate from a% to b% at $\varepsilon = 0.05$, at a throughput cost of c%.

Appendix I. Network-IDS Transfer (Not in Headline Claims)

Transfer F1 to CSE-CIC-IDS2018/CIC-IoT2023/UNSW-NB15 after fine-tuning only the head: a/b/c, median over 5 seeds. These numbers are not used in the headline claims because the datasets do not represent agent-protocol threats.

Appendix J. Ethics Considerations

The adaptive-attack suite in §8 contains code for E1–E5 that could in principle be adapted to attack production MCP servers. We release the scaffolding needed to reproduce the paper but not working exploits against live third-party services. All protocol traces are synthetic or from public benchmarks, and no live third-party agent deployment was targeted. We do not include CVE identifiers in the submission. [If the E6 red team is run: state the ethics approval and participant terms. If any new protocol flaw is found: state that it was reported under coordinated disclosure. Otherwise delete these clauses.]

Appendix K. Generative AI Usage

[Name the LLMs actually used to generate the A2A-sandbox traces and for the E3 attacker.] Prompts, seeds and API response caches are released in the artifact, within the limits that closed-model drift imposes on reproducibility. Generative AI was used for editorial purposes in this manuscript, and all outputs were inspected by the authors to ensure accuracy and originality.

Appendix L. Artifact Availability

An anonymised artifact is available at https://anonymous.4open.science/r/⟨ID⟩. It contains the detector, the smoothing, sparse-smoothing and composed certificates, the Hedge controller, the adaptive attacks, the training and evaluation harness, all configuration files, and 35 tests. `scripts/reproduce_all.sh` runs the tests and regenerates every synthetic-input result in about ten minutes on a CPU; `scripts/reproduce_all.sh --full` additionally reruns the benchmark experiments on one GPU. Recorded outputs are in `reproducibility/experiments_sp2027.json`. The artifact is frozen at the submission deadline and is not updated during review, per the S&P 2027 artifact policy.